Computer Science Internal Assessment: Criterion A

Total Word Count: 431

Defining The Problem:

My client, **Mr.Brown**, is a school's IB Computer Science teacher. He is responsible for teaching IB Higher Level and Standard Level Computer Science and Pre-IB Computer Science. As a Computer Science teacher, he wants his students to have coding exercises to improve their logical thinking skills and overall coding. With this, my client utilizes a website called CodingBat, where his students can create an account, join my client's class, solve a range of Java and Python questions, and have my client view their progress throughout each question. However, he finds that he is unable to modify testing parameters, suggest challenges for students who find certain questions easy, and input his own custom questions for his students. Additionally, he does not appreciate how students can easily click the 'Show Solution' button to answer a question rather than working on a solution independently.

Currently, my client finds CodingBat to be "a useful resource"; however, he would prefer a system that allows him personalization to his preferences for his students and has a more "aesthetically pleasing". With this, my client requested for me to create a version of CodingBat that would allow him to add and modify questions and their testing parameters, challenge his students with more difficult questions if they are doing well, or provide them with easier questions if they are struggling with coding solutions, and ensure that his students are not cheating by immediately looking at the solution GUI (See Appendix A-1).

Rationale for Proposed Solution:

To create this program, I will use Java rather than Python, as it offers better performance due to its compiled nature, crucial for handling multiple test cases and maintaining responsive GUI interactions. Additionally, Java's object-oriented nature enhances code efficiency, reusability, and security through encapsulation, while its platform independence ensures system accessibility across all operating systems. I will be using a database instead of Google Excel as the increase in student data or the number of questions my client adds to the database will be able to handle the data of 200 students per year efficiently and, additionally, those who have graduated. However, Excel would boggle down when handling mass amounts of data. Furthermore, databases offer more reliable data validation, integrity, and security than Excel. I will use SQLite as my database management system because it allows students to have the most up-to-date questions from my client and provides online access to the database. Additionally, I will be using Jython and JDK 17 to execute student solutions, as these libraries enable secure and efficient runtime evaluation of both Python and Java code while maintaining system integrity through controlled execution environments.

Success Criteria:

1. Login System
    a. Incorrect Login information will not allow the user to access the system
    b. Correct Login information will allow the user to access the system
    c. Allows for incorrect data to be handled by having validation rules
    d. Students can use their personal login information to access the student-accessible portion of the system.
    e. Teachers will be able to use their personal login information to access their students' progress and alter questions (the teacher-accessible portion of the system)
2. User Interface
    a. Student
        i. Display a choice of Java and Python Questions:
            - Provides a user-friendly interface where students can select between Java and Python programming languages.
        ii. Test student answers and display correct and wrong outcomes based on different test cases of the question
        iii. Recommends questions to Students that depend on the difficulty of the subsequent question based on previous correct and incorrect attempts
        iv. Show personal progress of current question:
            - Displays students' progress through visual progress bars
        v. Students are only shown solution hints of questions after 3 failed attempts

    b. Teacher

i. Dashboard for question management:
   - Provides a centralized dashboard where teachers can manage questions in the database, including adding, editing, and deleting questions.

ii. Dashboard for student management:
   - Provides a centralized dashboard where teachers can manage students, including editing student information, and deleting students.
   - Allows teachers to monitor the performance of individual students based on viewing past attempts and viewing their total number of correct and incorrect question attempts

iii. Dashboard for class management:
   - Provides a centralized dashboard where teachers can manage classes in the database, including adding, editing, and deleting classes, as well as adding students to a specific class.